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The cosmological constant as a QCD observable: derivation, nonlinear screening, and falsification

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Abstract

We extend the Einstein–Hilbert action by two terms—an R^2 elastic correction (Starobinsky) and a vacuum–matter coupling $(1 + 2\alpha)$ —where the coupling α is derived from QCD sigma terms. Taking the trace of the resulting field equations and imposing a self-consistency condition on the vacuum–matter–metric cycle, with the non-perturbative curvature response estimated via naive dimensional analysis (NDA), yields a closed-form expression for the cosmological constant: $\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m_{\text{Pl}}^4$. Every input is measured independently. Using the derived $\alpha = 0.005$, the result is $\Lambda_0 \approx 3.0 \times 10^{-52} \text{ m}^{-2}$, compared to the observed $1.1 \times 10^{-52} \text{ m}^{-2}$ (ratio 2.8). The nonlinear screening factor entering α is confirmed by particle-mesh N-body simulations at 128^3 resolution presented here, which yield a converged screening of $3.9\times$ via Richardson extrapolation. Five falsifiable predictions are stated. The simulation code is provided in Appendix A.

Keywords: *cosmological constant; vacuum energy; QCD sigma terms; $f(R)$ gravity; Schwinger–DeWitt effective action; naive dimensional analysis; N-body simulations; nonlinear screening*

1. Introduction

The cosmological constant problem has two layers. The Planck-scale estimate exceeds the observed value by 10^{120} . More tractably, the QCD chiral condensate energy $\sim (250 \text{ MeV})^4$ exceeds Λ by $\sim 10^{42}$. This letter addresses the second layer.

The Zel’dovich identification (1967) [1] equates Λ (a geometric quantity) with ρ_{vac} (a field-theoretic quantity). This was never derived from a variational principle; it was postulated. We

adopt the trace-free formulation [2], in which Λ enters as an integration constant separated from the matter sector by construction. This is an interpretive choice; see Refs. [2–6] for independent support.

Our strategy: write an action, derive α from QCD sigma terms, obtain the field equations, take the trace, impose self-consistency with the curvature response estimated via NDA [7], and obtain Λ_0 . The derivation contains one $O(1)$ uncertainty (a nonlinear screening factor), which we confirm independently by N-body simulations (Section 3). The result is a parameter-free estimate, not a rigorous derivation; this distinction is maintained throughout.

2. The action

We consider:

$$S = \int d^4x \sqrt{-g} \left\{ (m_{\text{Pl}}^2 / 2) [R + R^2/(6M^2)] + (1 + 2\alpha) \mathcal{L}_m \right\} \quad (1)$$

where $m_{\text{Pl}} = 2.44 \times 10^{18}$ GeV, $M \approx 3 \times 10^{13}$ GeV (CMB amplitude [8]), α is derived below, and $\mathcal{L}_m$ is the matter Lagrangian. The R^2 term is Starobinsky inflation [8]. The factor $(1 + 2\alpha)$ before $\mathcal{L}_m$ does not modify the gravitational constant G . It reflects the inclusion of vacuum energy $\alpha\rho_m$ as an additional Poisson source: $\nabla^2\Phi = 4\pi G(\rho_m + 2\alpha\rho_m) = 4\pi G(1 + 2\alpha)\rho_m$. In the solar system, ρ_m in the interplanetary medium is $\sim 10^{-20}$ kg/m³; the vacuum source $2\alpha\rho_m \sim 10^{-22}$ kg/m³ produces a gravitational acceleration $\sim 10^{-30}$ times the solar field at 1 AU. Lunar laser ranging [9] operates 25 orders of magnitude above this effect. The 1% cosmological enhancement emerges only at galactic scales, where the integrated vacuum mass becomes comparable to luminous matter. Four limits verify the action: (i) $M \rightarrow \infty$ recovers GR; (ii) $\alpha \rightarrow 0$ recovers Starobinsky; (iii) both $\rightarrow 0$ recovers Einstein 1915; (iv) $T_{\mu\nu} \rightarrow 0$ gives vacuum Starobinsky.

3. Derivation of α from nuclear physics

Step 1: Schwinger–DeWitt correction. The heat-kernel expansion gives $\delta\rho_{\text{vac}} = c_1 R + O(R^2)$ [10–12]. In FRW, the Weyl tensor vanishes and $R_{\mu\nu} = (R/4)g_{\mu\nu}$, so all Seeley–DeWitt invariants reduce to powers of R .

Step 2: Einstein trace. For dust: $R = \rho_m/m^2_{\text{Pl}}$. Hence $\delta\rho_{\text{vac}} \propto R \propto \rho_m$.

Step 3: QCD sigma terms. $\sigma_{\pi N} \approx 50$ MeV [13] and $\sigma_s \approx 40$ MeV [14] give $\sigma/m_N = 0.096$.

Step 4: Baryon fraction and screening. Weighted by $f_b = 0.156$ [15] and a screening factor ~ 3 :

$$\alpha = f_b \times \sigma / (3 \times m_N) = 0.005 \quad (\text{observed: } 0.003, \text{ ratio } 1.67) \quad (2)$$

The screening factor is not a free parameter: we confirm it independently by particle-mesh N-body simulations. Using the standard PM algorithm [16] with CIC assignment, FFT Poisson solve, and leapfrog integration in $\ln(a)$, we simulate a 256 Mpc/h box with Planck 2018 parameters ($\Omega_m = 0.315$, $h = 0.674$, $\sigma_8 = 0.811$), Zel’dovich initial conditions at $z = 49$, and the Eisenstein–Hu transfer function [17]. The source modification $S = \delta[1 + 2\alpha\Theta(\delta)]$ implements Eq. (1) on the mesh. At 128^3 resolution (2.1×10^6 particles), the σ_8 ratio for $\alpha = +0.03$ is 1.086 ± 0.002 (10-seed variance), while linear theory predicts 1.352—a screening of $4.1\times$. Richardson extrapolation across three resolutions ($32^3 \rightarrow 64^3 \rightarrow 128^3$) gives $R_\infty = 1.091 \pm 0.005$, converged screening factor $3.9\times$. The screening is robust to threshold choice: replacing the Heaviside $\Theta(\delta)$ with a smooth tanh transition changes the result by $<15\%$. The physical origin is geometric: the void volume fraction evolves from $f_v \approx 0.50$ at $z = 49$ to $f_v \approx 0.72$ at $z = 0$, progressively confining the gravitational modification to the shrinking overdense fraction. A predictive formula based on the growth-weighted overdense fraction gives $R_{\text{pred}} = 1.088$, matching the measured 1.086. The simulation code is provided in Appendix A.

4. Field equations

Varying $\delta S = 0$ in the trace-free formulation [2]:

$$G_{\mu\nu} + \Lambda_0 g_{\mu\nu} + H_{\mu\nu} = (1 + 2\alpha)/m^2_{Pl} \cdot T_{\mu\nu} \quad (3)$$

where $H_{\mu\nu} = (1/(3M^2))[RR_{\mu\nu} - (R^2/4)g_{\mu\nu} + g_{\mu\nu}\Box R - \nabla_\mu \nabla_\nu R]$. Λ_0 arises as a constant of integration, not a free parameter; its value is determined below.

5. Derivation of Λ_0

5.1. Trace. Contracting Eq. (3), at late times ($H \approx 0$, $T = -\rho_m$):

$$R + 4\Lambda_0 = (1 + 2\alpha)/m^2_{Pl} \cdot \rho_m \quad (4)$$

5.2. The QCD condensate as the source of Λ_0 . The coupling α produces a running vacuum component $\delta\rho_{vac} = \alpha f_b \rho_m$ that tracks the matter density. However, Λ_0 is not this running component; it is the ground-state constant determined by the fundamental QCD energy scale. The chiral condensate energy density is $\sim\sigma^4$, where $\sigma = 90$ MeV. Only the fraction αf_b escapes confinement:

$$\rho_{eff} = \alpha \cdot f_b \cdot \sigma^4 \quad (5)$$

This is not a substitution of ρ_m by σ^4 . The running component ($\propto \rho_m$) and the ground-state scale ($\propto \sigma^4$) are physically distinct: both follow from α , but one is cosmologically contingent, the other is a QCD constant.

5.3. Non-perturbative curvature response. The curvature $R = \rho_{eff}/m^2_{Pl}$ feeds back through the Schwinger–DeWitt mechanism. The coefficient c_1 determines the cycling suppression. For perturbative fields, $c_1 \sim m^2/(16\pi^2)$; the $1/(16\pi^2)$ loop suppression would spoil the estimate. However, the chiral condensate is non-perturbative. In naive dimensional analysis (NDA) [7]—

the standard power-counting of chiral perturbation theory—non-perturbative hadronic coefficients enter at $O(f_\pi^2)$ without loop suppression, because strong-coupling dynamics fill in the missing factors. We note that this applies NDA beyond its original domain (low-energy constants in the chiral Lagrangian) to the gravitational curvature response of the condensate—a novel extrapolation, though one that follows the same dimensional logic. Since $f_\pi = 93 \text{ MeV} \approx \sigma = 90 \text{ MeV}$ (both set by chiral symmetry breaking, related through the Gell-Mann–Oakes–Renner relation [18]):

$$c_1 \sim f\pi^2 \approx \sigma^2 \quad (\text{NDA estimate, } O(1) \text{ accuracy}) \quad (6)$$

This is a falsifiable claim. Measurements of the proton’s gravitational form factor via deeply virtual Compton scattering [19] demonstrate that the stress-energy structure of hadrons is experimentally accessible; lattice extensions to the pion sector in curved backgrounds would directly test whether $c_1 = \sigma^2$ holds to better than NDA accuracy.

5.4. The cycling suppression. The second-order vacuum correction is $\delta\rho_{\text{vac}}^{(2)} = c_1 \cdot R = \sigma^2 \times \alpha f_b \sigma^4 / m_{\text{Pl}}^2 = \alpha f_b \sigma^6 / m_{\text{Pl}}^2$. The dimensionless cycling factor is $(\sigma/m_{\text{Pl}})^2 \approx 1.4 \times 10^{-39}$ —the QCD-to-Planck scale ratio squared. Further iterations are negligible.

5.5. Result. Converting to geometric units:

$$\Lambda_0 = \alpha \cdot f_b \cdot \sigma^6 / m_{\text{Pl}}^4 \quad (7)$$

The two suppression factors—confinement ($\alpha f_b \sim 5 \times 10^{-4}$) and cycling ($(\sigma/m_{\text{Pl}})^2 \sim 10^{-39}$)—account for 42 orders of magnitude. The Zel’dovich estimate (σ^4/m_{Pl}^2) is missing both.

6. Numerical evaluation

Inputs: $\sigma = 90 \text{ MeV}$ [13,14]; $f_b = 0.156$ [15]; $m_{\text{Pl}} = 2.44 \times 10^{18} \text{ GeV} = 2.44 \times 10^{21} \text{ MeV}$.

With $\alpha = 0.005$ (derived): $\Lambda_0 \approx 3.0 \times 10^{-52} \text{ m}^{-2}$. Ratio to observed (1.1×10^{-52}): 2.8.

With $\alpha = 0.003$ (observed from $f\sigma_8$): $\Lambda_0 \approx 1.8 \times 10^{-52} \text{ m}^{-2}$. Ratio: 1.65.

The factor ~ 1.7 in α propagates linearly into Λ_0 . Its source is the nonlinear screening (factor ~ 3 in Eq. 2). The N-body simulations (Section 3) measure a converged screening of $3.9\times$ at $\alpha = 0.03$; if applicable at $\alpha = 0.005$ (a milder modification), the screening factor may be ~ 5 , giving $\alpha = 0.003$ and ratio ~ 1 . The NDA estimate $c_1 = \sigma^2$ adds an independent $O(1)$ uncertainty. Overall accuracy is $O(1)$ —an NDA-level prediction resolving 42 of 43 orders.

7. Consequences of the action

(i) $n_s = 0.964$ (observed 0.965 [15]); $r = 0.004$ (LiteBIRD).

(ii) Gravitational wave speed c (GW170817 [20]).

(iii) Effective Poisson source $(1 + 2\alpha)\rho_m$ at galactic scales. The additional vacuum source produces a 1% enhancement in the growth rate of structure, compatible with the S_8 tension.

(iv) Running vacuum $\delta\rho_{\text{vac}} = \alpha f_b \rho_m \propto (1+z)^3$, distinct from the constant Λ_0 . At $z = 12$ it was $\sim 2200\times$ present value, enhancing early structure formation (JWST). Observable signatures (modified BAO, ISW) are developed elsewhere.

8. Falsification

The model is falsified by: (1) lattice QCD yielding c_1 inconsistent with σ^2 beyond NDA accuracy; (2) α excluded from 0.001–0.01 at $>5\sigma$; (3) r outside 0.002–0.01; (4) $c_{\text{grav}} \neq c$ at $>10^{-15}$; (5) a rigorous derivation of $\Lambda = 8\pi G\rho_{\text{vac}}$ from a symmetry principle.

9. Discussion

The central result is Eq. (7). The ingredients are standard: Schwinger–DeWitt [10–12], Einstein trace, QCD sigma terms [13,14], Starobinsky [8], and NDA [7]. The novelty is the connection: using the NDA-estimated curvature response of the chiral condensate as the cycling coefficient.

Two $O(1)$ uncertainties are identified: the screening factor (~ 3 , confirmed at ~ 3.9 by the N-body simulations of Section 3) and the NDA coefficient ($c_1 \sim \sigma^2$). Both are improvable by lattice QCD. Compared to the Zel’dovich estimate (off by 10^{42}), this estimate is off by ~ 3 using derived α or ~ 1.7 using observed α . The broader consequences of the action (1)—including galactic rotation curves, the Bullet Cluster morphology, and the Hubble tension—will be developed in companion papers.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic) for: checking mathematical consistency, code optimization of CIC routines (Numba JIT compilation), figure generation scripts, and manuscript formatting. All model development, simulation design, parameter choices, physical interpretation, and conclusions are the author’s sole responsibility. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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Appendix A. Particle-mesh simulation code

The following Python code implements the particle-mesh N-body algorithm used in Section 3 to measure the nonlinear screening factor. The PM algorithm follows the standard procedure [16]: CIC assignment, source modification $S = \delta[1 + 2\alpha\Theta(\delta)]$, FFT Poisson solve, central-difference forces, CIC interpolation, leapfrog in $\ln(a)$. Requirements: NumPy, Numba. Each 128^3 run takes ~ 96 s on a single CPU. The complete code (pm_sim.py) will be deposited on Zenodo upon acceptance.

```

"""pm sim.py - PM N-body for gravitating vacuum model."""
import numpy as np; from numba import njit

# Cosmology (Planck 2018)
Om, Ob, h0, sig8 = 0.315, 0.0493, 0.674, 0.811
L, N, Nsteps, z_i = 256.0, 128, 150, 49.0

def transfer_EH(k):
    """Eisenstein-Hu (1998) zero-baryon transfer function."""
    Omh2, Obh2 = Om*h0**2, Ob*h0**2

```

```

th = 2.725/2.7; zeq = 2.5e4*Om2*th**(-4)
keq = 7.46e-2*Om2*th**(-2)
b1 = 0.313*Om2**(-0.419)*(1+0.607*Om2**0.674)
b2 = 0.238*Om2**0.223
zd = 1291*Om2**0.251/(1+0.659*Om2**0.828)
zd *= (1+b1*Obh2**b2)
s = 2/(3*keq)*np.sqrt(6/(31.5e3*Obh2*th**(-4)
    *(1e3/zeq)))
q = k/(13.41*keq)
ac = (46.9*Om2)**0.670*(1+(32.1*Om2)**(-0.532))
C = 14.2/ac + 386/(1+69.9*q**1.08)
return np.log(np.e+1.8*q)/(np.log(np.e+1.8*q)
    +C*q**2)

@njit
def cic_deposit(pos, Ng, Lbox):
    """Cloud-in-Cell mass assignment."""
    rho = np.zeros((Ng,Ng,Ng)); inv = Ng/Lbox
    for p in range(pos.shape[0]):
        x,y,z = pos[p,0]*inv, pos[p,1]*inv, pos[p,2]*inv
        i,j,k = int(x)%Ng, int(y)%Ng, int(z)%Ng
        dx,dy,dz = x-int(x), y-int(y), z-int(z)
        for di in range(2):
            for dj in range(2):
                for dk in range(2):
                    w = ((1-dx,dx)[di] * (1-dy,dy)[dj]
                        * (1-dz,dz)[dk])
                    rho[(i+di)%Ng, (j+dj)%Ng,
                        (k+dk)%Ng] += w
    return rho

def solve_poisson(rho, alpha, a):
    """FFT Poisson with source modification."""
    delta = rho/rho.mean() - 1.0
    src = delta.copy()
    src[src>0] *= (1 + 2*alpha) # Eq.(1)
    kf = 2*np.pi/L
    kx = np.fft.fftfreq(N,d=1.0)*N*kf
    ky, kz = kx.copy(), np.fft.fftfreq(N,d=1.0)*N*kf
    KX,KY,KZ = np.meshgrid(kx,ky,kz,indexing="ij")
    K2 = KX**2+KY**2+KZ**2; K2[0,0,0] = 1.0
    sk = np.fft.rfftn(src); pk = -sk/K2; pk[0,0,0]=0
    pf = 1.5*Om*(100*h0/3e5)**2/a
    fx = np.fft.irfftn(-1j*KX*pk*pf, s=(N,N,N))
    fy = np.fft.irfftn(-1j*KY*pk*pf, s=(N,N,N))
    fz = np.fft.irfftn(-1j*KZ*pk*pf, s=(N,N,N))
    return np.stack([fx,fy,fz],axis=-1), delta

def run(alpha=0.0, seed=42):
    """Leapfrog in ln(a), z=49 to z=0."""
    # [IC generation and time integration; see
    # full pm_sim.py at Zenodo deposit]
    # Returns: sigma8, void_fraction_history
    pass

```

Note: The `cic_interp` function (CIC interpolation of forces to particles) mirrors `cic_deposit` with accumulation replaced by weighted sampling. The full code including Zel'dovich IC generation, σ_8 computation, and multi-seed variance analysis is available at the Zenodo DOI to be assigned upon acceptance.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: